

Conrado Aguilar

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EDUCATION

Toronto Metropolitan University (TMU)

Toronto, ON

Bachelor of Engineering in Software Engineering (Co-op), September 2023 – Expected May 2028

- **Relevant Courses:** Data Structures & Advanced Algorithms, Object-Oriented Programming, Signals and Systems, Microprocessor Systems.

EXPERIENCE

Sports Instructor

Mississauga, ON

Lifetime Fitness - May 2024 - Present

- Tracked participation and performance using **structured lesson plans while prioritizing safety-first protocols**, resulting in improved engagement, confidence, and teamwork among 30+ children per session.
- Managed classroom behavior through **positive reinforcement and conflict-resolution techniques**, leading to a safe, inclusive environment for children of diverse abilities and backgrounds.

Director of Marketing

Toronto, ON

IEEE Toronto Metropolitan University, September 2025 - Present

- Led content strategy, creation, and scheduling across social media platforms as part of the IEEE TMU marketing team, ensuring consistent brand voice and posting cadence
- Generated **160K+ content views** and grew **follower base by 200+** in a single month (September 2025) through targeted content planning
- Designed and executed brand-aligned campaigns that boosted student engagement and expanded overall reach across platforms.

Planning Committee Member

Toronto, ON

Metropolitan Undergraduate Engineering Society, September 2023 - May 2024

- Planned and set up large-scale MUES events using **team-based logistics and structured task lists**, resulting in entertainment events for hundreds of engineering students.
- Brainstormed event concepts and coordinated with other committees using **collaborative planning sessions** while remaining aware of budget complications.

PROJECTS

Personal Portfolio Website

React | HTML | TypeScript | CSS | Vercel

- Built a responsive personal portfolio website primarily using **React, CSS and TypeScript** enabling a professional platform that showcases my engineering projects, skills, and academic experience to recruiters.
- Designed a modular UI using **component-based architecture, props, and state**, allowing sections such as projects, contact, and about pages to scale cleanly as content grows.
- Deployed and maintained the site using **Vercel and Git-based workflows** (branching, pull requests, code reviews), ensuring a continuously updated, in alignment with industry practices.

Ticket Prices Web Scraper

Python | JavaScript | Flask | API | Automation

- Built a backend data aggregation system using **Python** to collect and process data from multiple sources, enabling efficient comparison of large datasets across platforms.
- Designed a modular backend architecture with structured classes and request handling, enabling scalable data pipelines and easier system maintenance.
- Implemented scheduled scraping **using automation scripts**, resulting in up-to-date pricing without manual user input.
- Integrated a frontend interface using **JavaScript and HTML/CSS and Vercel for deployment** resulting in an accessible tool for comparing ticket prices in real time.

CPU Design

Quartus II | VHDL | FPGA Simulation | Digital Logic

- Designed and implemented an 8-bit CPU using **VHDL and modular digital logic** with an Arithmetic Logic Unit capable of executing arithmetic and logic operations.
- Implemented instructions through **a finite state machine (FSM)** resulting in reliable sequencing and coordinated processor operations.
- Tested and debugged hardware behavior using **Quartus II simulations and FPGA workflows**, resulting in validated logic paths and stable processor execution.

Market Intelligence Agent

Python | TypeScript | LLMs | API | Automation

- Developed a multi-agent AI system using **Python and FastAPI** by orchestrating specialized agents (research, extraction, analysis, reporting), enabling automated generation of structured market intelligence reports from natural language inputs.
- Integrated multiple **APIs** and data sources to build reliable data pipelines, improving data consistency and system performance.
- Designed and implemented a secure and scalable backend architecture using **FastAPI** and **Next.js**, ensuring reliable handling of long-running processes and protected system access.

Restaurant Application

React Native | App Development | Node.js | Expo Go | Supabase

- Collaborated with partners to build and launch a cross-platform Tinder-style restaurant application for 20+ beta users using **React Native** and **Node.js**, integrating REST APIs and **Supabase (PostgreSQL)**.
- Implemented a sign up and gated sign-in flows using **TypeScript** and **Node.js** backend logic, and integrated an **LLM** in the recommended layer, resulting in more personalized restaurant suggestions.
- Tested the application using **Expo Go**, enabling rapid iteration, device testing, and validation across **iOS** and **Android** during beta.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++, React, React Native, Node.js, VHDL, and HTML.

Frameworks: React, Node.js, SpringBoot, FastAPI, React Native, JavaFX, and JUnit.

Developer Tools: Git/Github, Docker, Quartus II, OpenAI API, VS Code, Cursor, Vercel, Replit, Supabase, Linux, Vercel and Expo.

Certifications: Introduction to Artificial Intelligence (IBM), Claude Certified Architect (In Progress).